

Threat Intelligence Correlation & Alert Prioritisation Assistant

From raw alerts to actionable intelligence

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CHALLENGE

Correlate multi-source alerts, identify real threats, map activity to MITRE ATT&CK, and generate concise BLUF summaries.



Security Teams Face Alert Overload

Volume

SIEM, EDR, IDS, firewall, WAF, proxy, and threat feeds generate too many alerts.

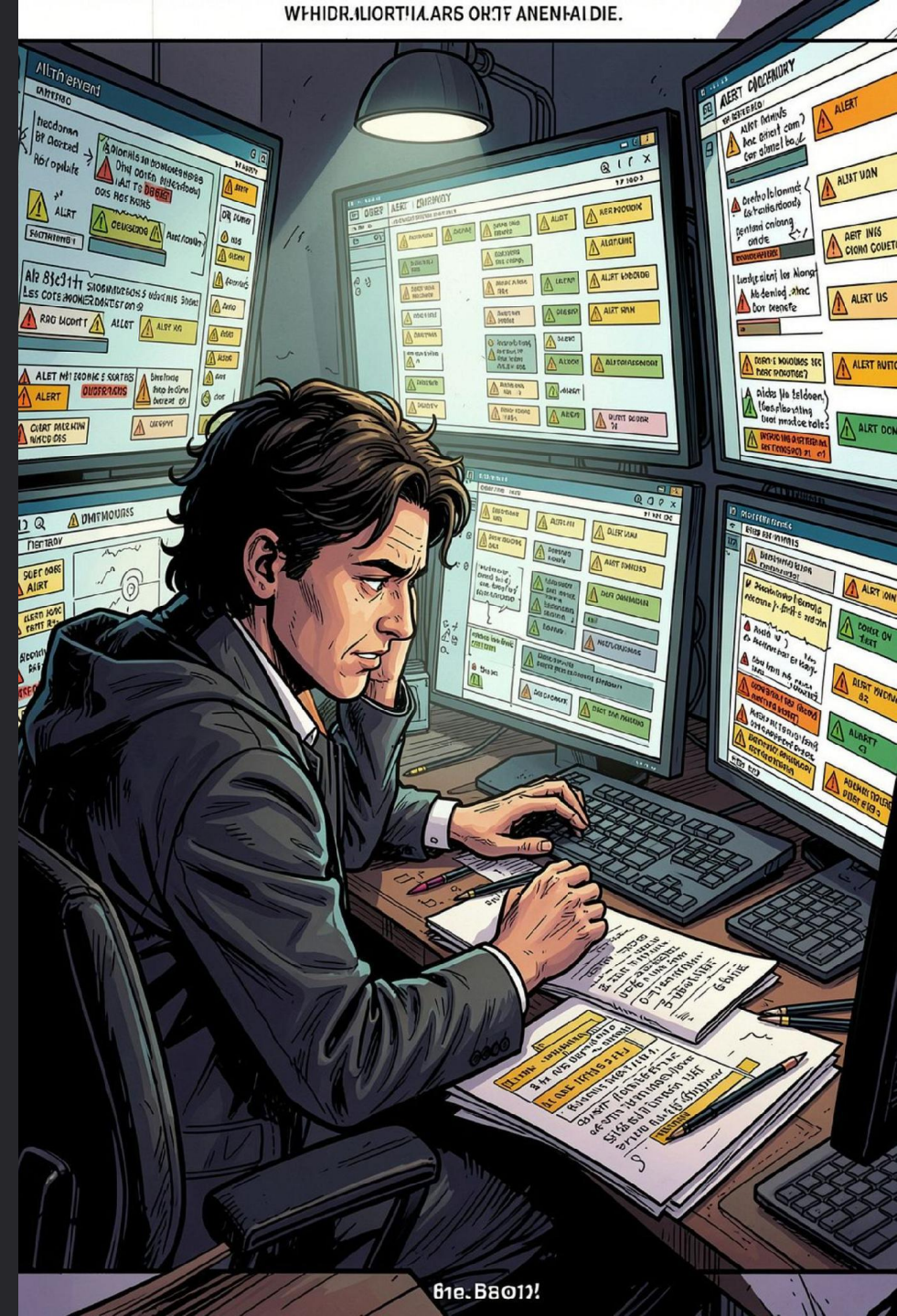
Noise

Duplicates and related events hide real attacks.

Manual Effort

Analysts spend too much time triaging logs instead of assessing threats.

- ❏ **Core Problem:** Turn scattered alerts into correlated incidents, prioritized threats, and actionable intelligence.



Proposed Solution: Correlation Engine

End-to-end alert prioritisation — from raw ingestion to commander-ready intelligence.



Ingest

Upload CSV alerts.



Correlate

Group by IP, host, time, behavior.



Prioritize

Score risk and rank by severity.



Map

Link to MITRE ATT&CK.



Summarize

Generate a BLUF brief.



Goal: Less noise, sharper focus, faster response.

System Architecture

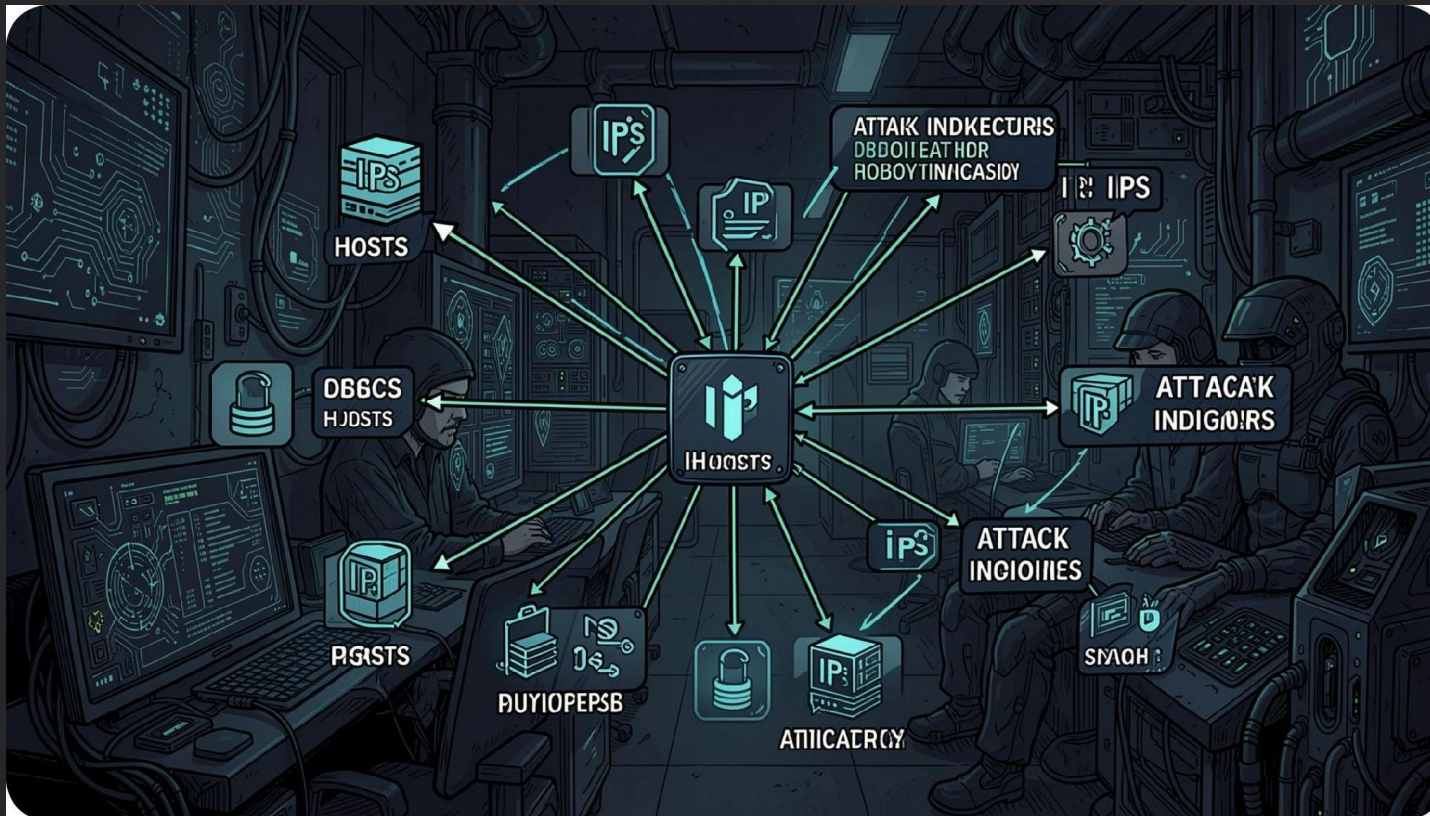


Input Sources

- SIEM / EDR
- IDS / Firewall
- WAF / Proxy
- Threat feeds

Raw alerts become prioritized intelligence for analysts and commanders.

Alert Correlation Engine



How Related Alerts Are Detected

Shared Indicators

IP, host, user, timestamp.

Behavior Match

Same tactic, similar malware, shared infrastructure.

Time Correlation

Events in the same attack window.

📌 **Example:** Port scan → C2 → malware → credential dump → ransomware = **one incident**

Risk Scoring & Prioritisation

INC-001 — Risk Score: 85 — HIGH

+40

High severity

+10

3 related alerts

+20

Ransomware

+15

C2 traffic

+10

Network scanning

+5

Multiple sources

Scoring Factors

- Severity
- Related alerts
- Malware
- C2 communication
- Scanning
- Credential theft
- Detection sources



Result: Critical threats surface first.

MITRE ATT&CK Mapping

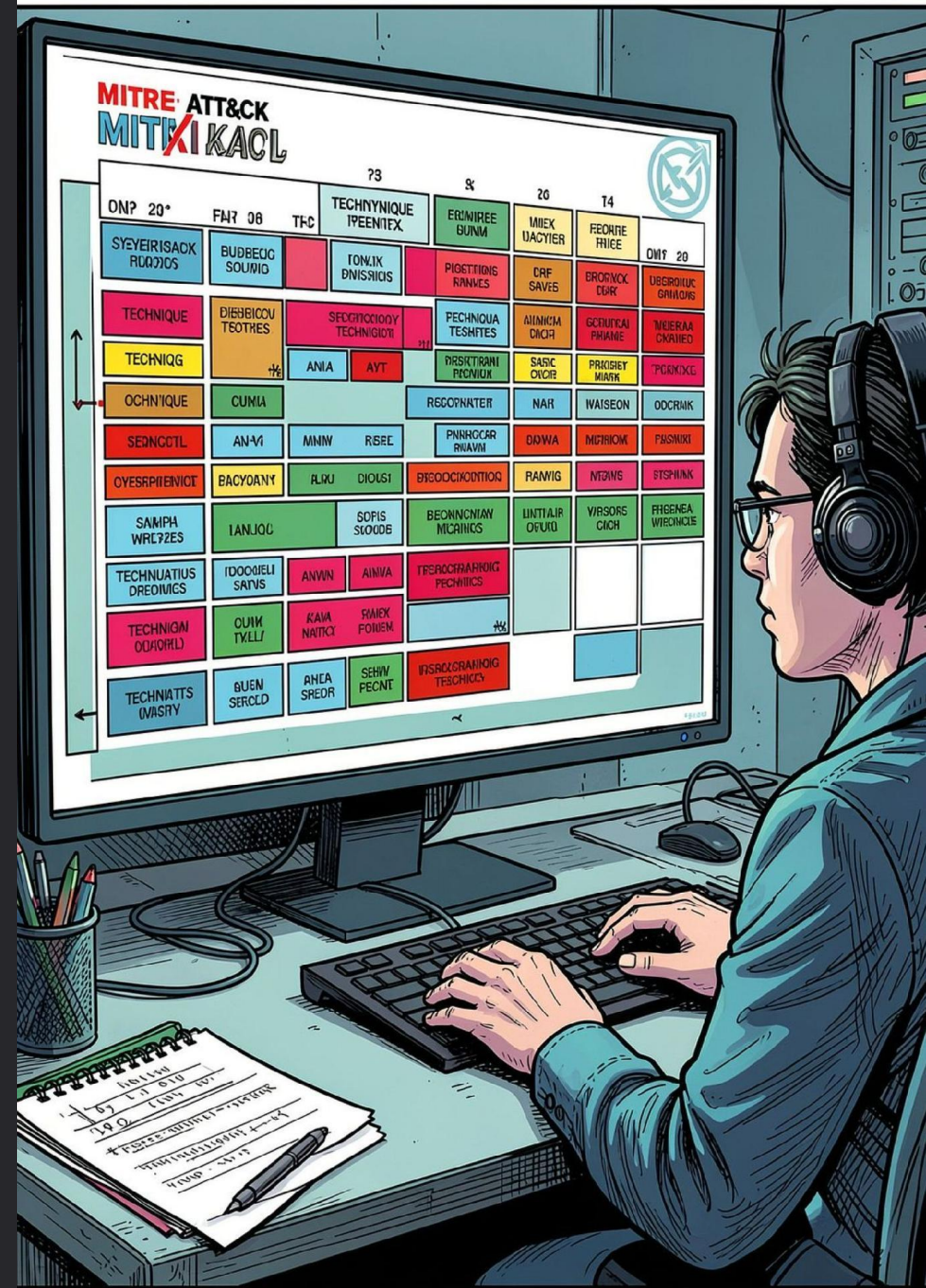
Behavior is mapped to standard adversary techniques for clearer investigation and threat hunting.

T1071 — App Layer Protocol
Command and Control

T1046 — Network Discovery
Discovery

T1486 — Data Encrypted
Impact

T1595 — Active Scanning
Reconnaissance



BLUF Intelligence Generation



Bottom Line Up Front

Turns raw alerts into a concise brief so commanders understand the threat in seconds.

Threat

Possible ransomware with C2 and reconnaissance activity.

Affected Asset

File server or workstation — **Critical / High**

Evidence

Port scan, C2 traffic, ransomware signature, malware activity.

MITRE Techniques

T1071, T1046, T1486, T1595

Recommended Action

Isolate host. Block infrastructure. Hunt for lateral movement and credential theft.

Prototype Results & Dashboard

20

Alerts Ingested

7

Correlated Incidents

1

Critical

3

High

3

Low

INC-001 Highlights

- Risk score: **85**
- 3 related alerts
- Ransomware + C2 activity
- MITRE mapping
- Intel brief generated

UI Capabilities

- CSV upload and correlation
- Prioritization and filtering
- Threat vs. false-positive classification
- Incident expansion and brief generation

Impact & Future Scope

For Analysts

Less fatigue, faster triage, better context.

For Commanders

Clearer picture, prioritized incidents, faster decisions.

Future Scope

- SIEM + STIX/TAXII integration
- IOC enrichment + anomaly detection
- SOAR-driven containment
- Historical campaign correlation

"From Alert Overload to Decision-Ready Intelligence."

